Public Economics and Development

Week 13: Recruiting Capable and Motivated Individual to Public Sector

Seung-hun Lee

Taipei School of Economics
National Tsing Hua University

May 19th, 2026

Why Selection Matters: The Stakes

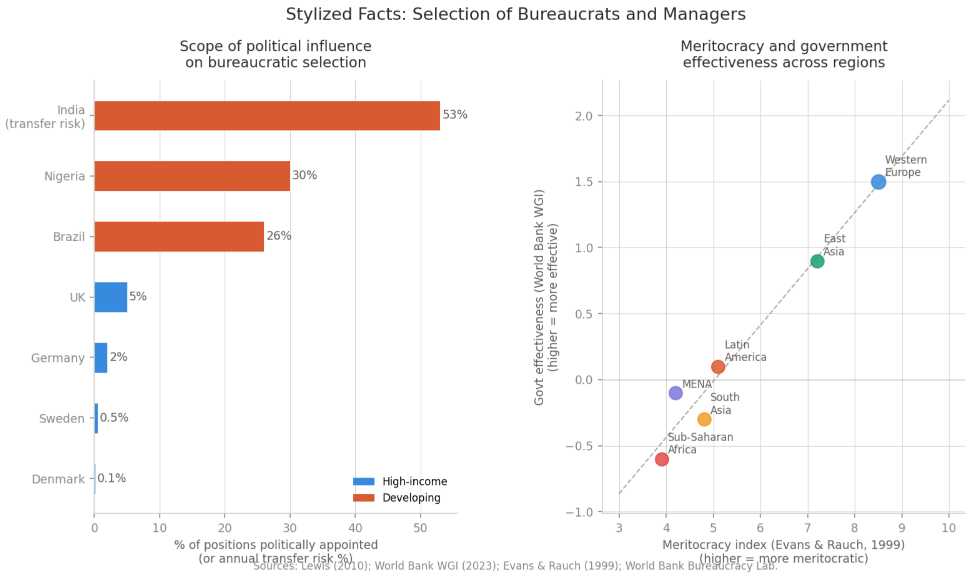
Patronage and meritocracy coexist in most countries

- Low-skilled positions: patronage common (reward political supporters)
- High-skilled positions: meritocracy more likely (incompetence could be costly)

Quality of governance strongly linked to **how** bureaucrats and personnel are selected

- But selection is often distorted — patronage and political connections pervasive
- Cost: Loyalty over capability — disrupts continuity and undermines service delivery
  - Junior bureaucrats under-invest in skills when transfer risk is high
  - Inconsistent and inefficient practices within public organizations
- Benefit: Discretion can weed out entrenched, low-performing workers
  - May be necessary when civil service protections shield underperformers
  - Allows principal to restore control over misaligned bureaucrats

Stylized facts on political influence in recruitment



What Determines Selection Quality?

Rule-based vs. discretionary selection — a fundamental tradeoff

- **Merit-based**: reduces favoritism, improves average quality, builds institutional trust
  - Civil service exams, competitive hiring, transparent criteria
  - But may not select for motivation, integrity, or adaptability
- **Discretion-based**: allows flexibility in matching candidates to roles
  - Can select for unobservable traits — motivation, integrity, cultural fit
  - But opens door to favoritism, patronage, and political interference

Financial incentives shape who applies — but not always in the right direction

- Higher wages attract more and better candidates
- May signal wrong job characteristics and crowd out intrinsically motivated candidates

Manager selection is often the missing link

- Managers determine practices within organizations
- Yet managerial positions often exempt from meritocratic reforms

Roadmap for today

1. Overview setup of the topic

2. Impact of management on the efficiency of public organizations
   2.1 Dal Bó, Finan, Folke, Persson, Rickne (2017): Who Becomes a Politician?
   2.2 Akhtari, Moreira, Trucco (2022): Political Turnover, Bureaucratic Turnover, and the Quality of Public Service
   2.3 Fenizia (2022): Managers and Productivity in Public Sector

3. Recruiting capable bureaucrats to public sectors
   3.1 Dal Bó, Finan, Rossi (2013): Strengthening State Capablities: The Role of Financial Incentives in the Call to Public Service
   3.2 Desserano (2019): Financial Incentives as Signals: Experimental Evidence from the Recruitment of Village Promoters in Uganda
   3.3 Moreira, Pérez (2024): Civil Service Exams and Organizational Performance: Evidence from the Pendleton Act

Dal Bó, Finan, Folke, Persson, Rickne (2017): Who Becomes a Politician?

Q: How to attract competent leaders while being representative?

Can democracies form competent and representative leadership (inclusive meritocracy)?

- Voters want to elect leaders capable of efficiently implementing policies
- Voters also want policy-makers who represent diverse interests
- Classical economic models suggest that these two are in a trade-off relation
  - Market wage < Returns from office for low-quality individuals (low opportunity cost)
  - Good politicians likely to ‘free-ride’ on bad politicians $\rightarrow$ Supply of good politicians $\Downarrow$
  - Even if good candidates can be selected, may make broad representation difficult

This paper uses data on Swedish local politicians to identify patterns in inclusive democracy

- Includes elected and *nonelected*: Address selection bias
- Detailed measures on quality and representation
- Identifies how both competent and representative leaders are selected in democracies

Context and data

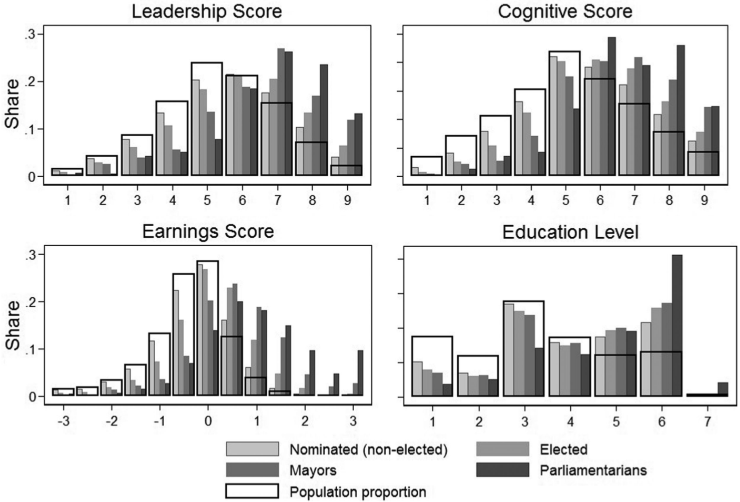
Swedish National MPs and Municipal council members

- MPs are full-time, while most municipal council members have other work
  - Mayors, selected among council members, are full-time and paid highly
- Low monetary cost to individuals running for local office
- Key motives: Intrinsic concerns with policy, social interaction in policy circles
- Left represents blue-collar, center-right represents white-collar workers

Data obtained from

- Sample: All candidates from 1980-2010
- Links candidates to admin registers (including military enlistment and tax records)
- Age, sex, education level, earnings, occupation, parental/sibling info, cognitive capacity

Positive selection of politicians based on abilities



- Beats classical prediction that there are more low-quality individuals entering politics

Are they socially representative?

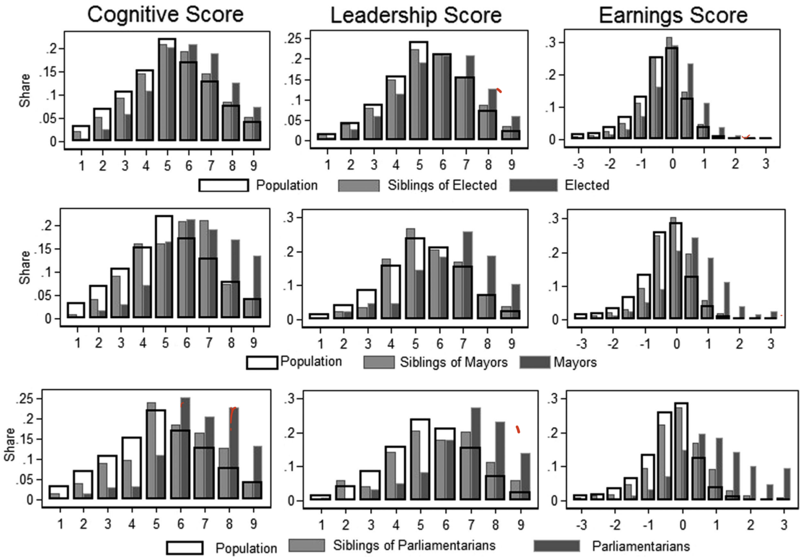
Determining source of positive selection

- Privileged access? Ability should matter little once elite status is controlled for
- Exclusive meritocracy? Key abilities are strongly associated with social background
- Inclusive meritocracy? High ability individuals come from all social backgrounds

Empirical tests

- Compare with siblings: Same "elitist background" $\rightarrow$ Positive selection still appears
- Parental income: Social background of politicians is evenly distributed
- Social class: Politicians from wide range of occupations represented

Elitism unlikely to explain positive selection



Neither does exclusive meritocracy

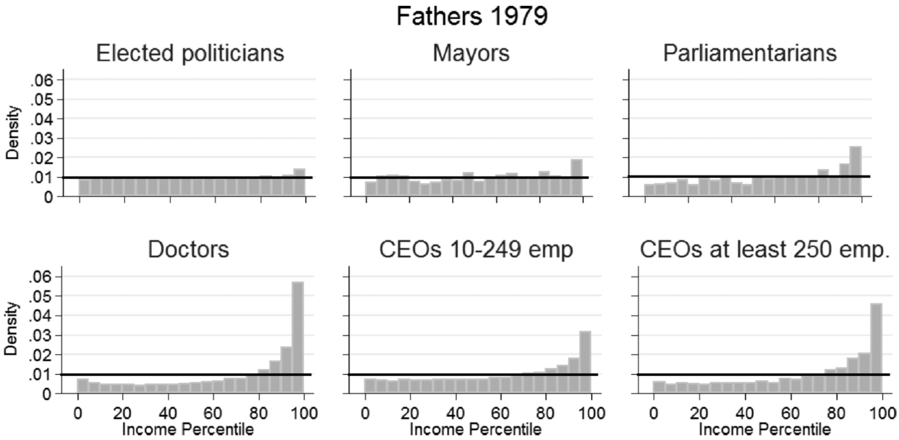


FIGURE IV

Distribution of Fathers of Politicians and Other Occupations across the Percentiles of Population Income

Discussion and takeaways

What drives competence and representation?

- Weak trade-off between competence and social background
- Positive selection on ability even among those with poorer backgrounds
- Intrinsic motives: Positive selection even for low-paid municipal councils
- Screening by parties: Sorts more able people to higher positions

Remaining questions

- When and where do these findings hold outside 1980-2010 Sweden?
  - Intrinsically motived individuals may not enter politics elsewhere, how to attract them?
  - Party incentives: Promote someone who can win vs. who has best policy
  - Other democratic countries? Sweden now?

Akhtari, Moreira, Trucco (2022): Political Turnover, Bureaucratic Turnover, and the Quality of Public Service

Q: How do changes in political leadership affect public service?

Countries vary in the extent to which politicians control appointment of personnel

- East Asia: Only the top chiefs and vice chiefs are politically appointed
- On the other end, much more positions are appointed by the president
  - Nearly all of top 500 government agency positions appointed in Brazil, Israel, Nigeria etc.
  - 26% of all govt personnel work under some threat of removal by the president
- Thus, politicaly motivated replacements are commonplace in the latter

This paper examines the effects of such scope of politically motivated appointments

- Leverage bureaucratic turnover following close elections in public schools
- Local politicians have significant influence over public school administration
- Turnovers in political leadership $\rightarrow$ New personnel $\rightarrow$ Education peformance $\downarrow$
- Documents trade-off between capability and loyalty

Municipality-run schools in Brazil

Municipal education in Brazil

- 5,565 decentralized municipalities are responsible for essential services
- Mayors are elected every 4 years, and can appoint $\sim$30% of workers via ‘contract’
    - Rest have passed civil service exams
- Municipalities responsible for daily operations and personnel of primary schools
    - 60% of headmasters in public schools are politically appointed
    - 35% of teachers hired by contract (rest passed civil service exams)

Why local politicians care about having loyalists in managerial positions

- Managers can use control over public programs and resources politically
- Headmaster position used as means to provide contracts to political supporters
- Thus, public positions used as means to consolidate political support

Data and empirical strategies

Data sources: Brazilian election and education data ('08-'12)

- Municipal personnel changes from Registry of Social Information (RAIS)
- National exam score (Prova Brasil) and School Census for edu quality and personnel
- Electoral data (party, vote share) from Tribunal Superior Eleitoral (TSE)
- Municipal characteristics from various geographical sources and political atlas

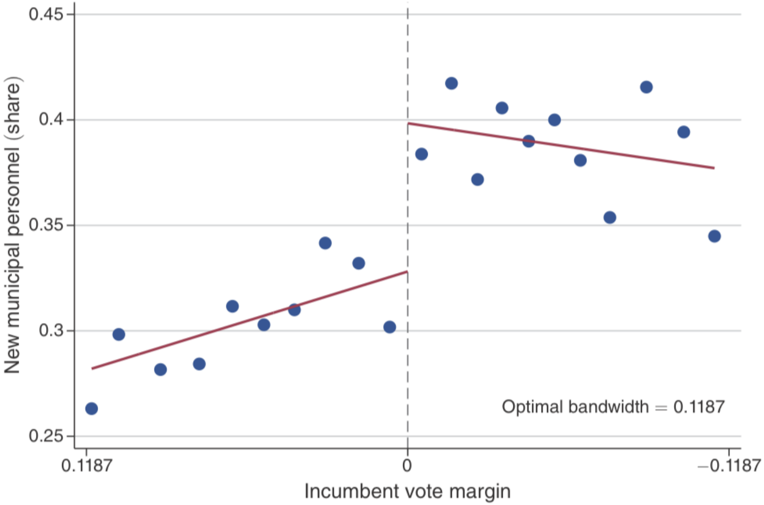
Using municipalities with close election results in regression discontinuity

- Not just comparing places with turnovers vs without one
  - Incompetent incumbent (bad public goods) $\rightarrow$ Voted out $\rightarrow$ reverse causality
- Narrow down to places with and without turnovers AND small electoral margins

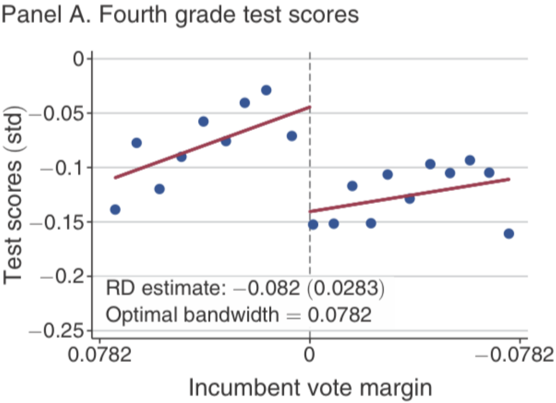
$$Y_{imt+1} = \alpha + \beta I[\text{Turnover}]_{mt} + \gamma \text{Margins}_{mt} + \delta I[\text{Turnover}]_{mt} \times \text{Margins}_{mt} + \lambda X_{jmt} + e_{jmt}$$

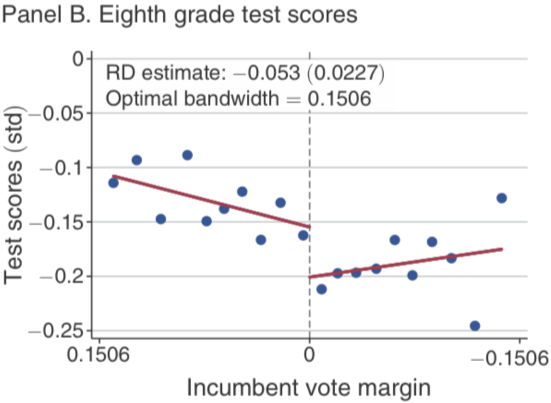
- Municipalities included in RD similar except for turnover, and no sorting on margins

Political turnover → Replacement of municipal personnel

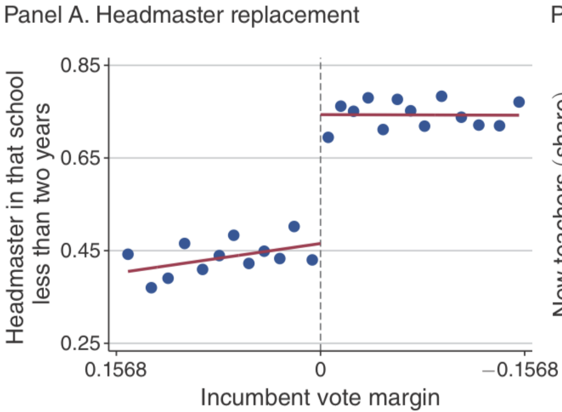


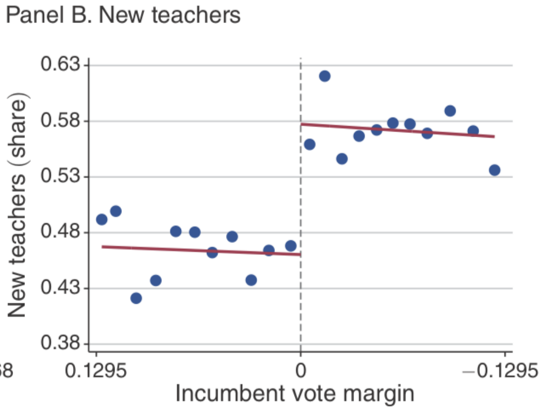
Lower educational performance of students





Drastic changes in school personnel (managerial and frontline)





Driving mechanisms and further questions

What explains (or does not explain) these changes?

- Possible confounders from municipality-wide forces? No effects on private schools
- No significant declines in educational spending in turnover municipalities
- Not related to political ideologies: Declines observed in both left/right wing parties
- Experience and capability of new personnel $\downarrow \rightarrow$ trade-off productivity and loyalty

What remains unanswered?

- Politically motivated replacements can be a costly byproduct on public services
  - Even in closely-contested, fair elections
- However, discretion may be needed to weed out low-performing, entrenched workers
- So what determines the relative strengths of these opposing forces?

Fenizia (2022):
Managers and Productivity in Public Sector

Q: Do talented managers in public sector improve performance?

How important are managers in public sectors?

- Oversee day-to-day operations and supervise policy implementation
- Limited tools available compared to private sector managers‘’
  - Limited discretion in firing and promoting workers
- How to measure performance of government agencies?

This paper uses innovative measures of performance to study the role of managers

- Italian Social Security Agency data (INPS): Covering varous welfare programs
- Uses welfare claims as measure of output (homogenous across different offices)
- All sites have same rules and physical capital $\rightarrow$ distinguish manager contributions

Background and data

Role of INPS

- Administers all social welfare and insurance programs in Italy
- Claims are processed in local offices run by single managers
- Managers allocate work, determine workplace practices, but cannot fire workers
- Managers selected among limited pool of candidates and rotate every 5 years

Data sources on managers (2011Q1 - 2017Q2)

- Output: $\sum$ claims weighed by complexity (procesing time) per worker
- Service quality: Timeliness and error rate
- Worker characteristics: Job title, age, office locations etc

Empirical strategies

Do productive managers improve performance?

- Fixed effects regression with office(i), time(t), and manager($m(i,t)$) fixed effects

$$\log P_{it} = \alpha_i + \tau_t + \theta_{m(i,t)} + u_{it}$$

- Requires that manager moves are exogenous conditional on office and time effects
  - Observable characteristics should not predict manager quality
  - Violated if better managers are assigned to worse offices
- Compare adjusted R^2 to test how much of output variation is explained by managers

What makes for a productive manager?

- Regress changes in outcomes with quarter before manager change to talent changes

$$y_{i,t+k} - y_{i,t-1} = \pi_0^k + \pi_1^k(\hat{\theta}_{i,new}^k - \hat{\theta}_{i,old}) + \gamma^k X_i + (e_{i,t+k} - e_{i,t-1})$$

Managers explain 10% of variation in productivity

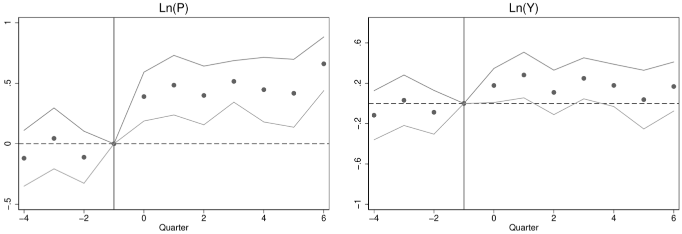
TABLE IV

ANALYSIS OF VARIANCE OF OFFICE PRODUCTIVITY

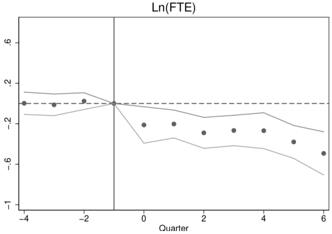
	(1) Ln(P)	(2) Ln(P)	(3) Ln(P)	(4) Ln(P)	(5) Ln(P)
N	3,316	3,316	3,316	3,316	3,316
R sq.	0.325	0.727	0.835	0.789	0.839
Adj. R sq.	0.324	0.679	0.762	0.720	0.765
Time FE	Yes	Yes	Yes	Yes	Yes
Office FE	No	Yes	Yes	No	No
Manager FE	No	No	Yes	Yes	No
Manag-by-Office FE	No	No	No	No	Yes
Pvalue			0.000	0.000	

- Adjusted R^2: 0.68 in (2) $\rightarrow$ 0.76 in (3): 8 p.p increase
- $0.08/0.762 \simeq 0.104$: Manager explains 10% of variation in productivity
- Robust to selective sorting, other ways of variance decomposition

Talented managers improve productivity, with workers decreasing...



(c)



Talented managers induced by retirement of older workers

k	(1) Retirement	(2) Separations	(3) Hires	(4) Fires	(5) Inbound T	(6) Outbound T
-4	0.041	0.099	0.178	0.003	0.120	-0.039
	(0.125)	(0.111)	(0.110)	(0.004)	(0.121)	(0.146)
-3	-0.044	-0.011	0.093	0.002	0.069	-0.111
	(0.088)	(0.078)	(0.090)	(0.004)	(0.090)	(0.133)
-2	-0.055	0.033	0.034	0.003	0.068	-0.122
	(0.059)	(0.053)	(0.073)	(0.004)	(0.082)	(0.112)
0	0.301	0.038	0.024	-0.008	0.031	-0.023
	(0.085)	(0.081)	(0.018)	(0.010)	(0.159)	(0.053)
1	0.393	-0.011	0.027	-0.056	-0.033	-0.010
	(0.100)	(0.099)	(0.033)	(0.031)	(0.163)	(0.067)
2	0.381	-0.057	0.024	-0.049	-0.196	-0.142
	(0.105)	(0.117)	(0.033)	(0.040)	(0.172)	(0.097)
3	0.392	-0.012	0.006	-0.063	-0.327	-0.234
	(0.117)	(0.126)	(0.038)	(0.045)	(0.174)	(0.114)
4	0.438	-0.0722	-0.015	-0.040	-0.385	-0.231
	(0.116)	(0.128)	(0.039)	(0.038)	(0.171)	(0.123)
5	0.413	0.015	0.005	-0.061	-0.403	-0.303
	(0.120)	(0.135)	(0.039)	(0.041)	(0.174)	(0.125)
6	0.399	-0.121	-0.082	-0.059	-0.537	-0.405
	(0.125)	(0.155)	(0.057)	(0.042)	(0.184)	(0.135)
N	318	318	318	318	318	318
Time FE	Yes	Yes	Yes	Yes	Yes	Yes

Takeaways and discussions

Other outcomes

- Trade-off in quality? Not observed
- Outcomes not driven by shifting towards easier claims to handle
- Thus managers do have a meaningful impact on productivity

Remaining questions

- The managers oversee relatively small teams with well-defined goals this context?
  - What about large projects and teams?
  - What about organization with no fixed roles and less structure?
- Other factors may determine managers' role in productivity
  - Ability to obtain inputs efficiently (no bribing, better communication)
  - Ability to attract more talented workers under the command

Dal Bó, Finan, Rossi (2013):
Strengthening State Capablities: The Role of Financial Incentives in the Call to Public Service

Q: Can fiscal incentives attract capable individuals to public sector?

Professionalized bureaucracy require individuals with capabilities and integrity

- These do not always co-move
  - Capable individuals may seek opportunities in private sectror (higher pay)
  - This may leave public sector with less able individuals
  - Not all capable individuals have integrity:
- So what makes up motivated agents and how to attract them to public sector?

This paper investigates role of incentives while investigating various aspects of quality

- How do monetary and non-monetary incentives attract candidates?
- Estimates if higher wages attract more candidates and those of higher quality
- Also estimates the labor supply elasticity facing public sector in Mexico
- Heterogeneity by locational characteristics (local crime rates)

Context of the experiment

Mexico's regional development program (RDP) in 2011

- Network of 350 agents & 50 coordinators to identify needs, report to federal govt
- Exogenously assign different wage offers across different posts (5000 vs 3750 MXP)
- Candidates screened to document various characteristics
  - Aptitude: intelligence (IQ) and other qualities known to determine market value
  - Motivation: Integrity, prosocial inclinations, public service motivation (interviews)
  - Attraction & commitment to policy making, justice, civicness, compassion, self-sacrifice
- Vacancies randomized to one of 4 strata $\rightarrow$ Offers randomized to indiv. in that stratum
  - High vs low wage, High (> 9) vs normal IQ (7-9 Ravens matrix)

Estimating causal effects: OLS (due to random assignment design)

- Effect on character Y in applicant pool (region): $Y_{icr} = \beta_1 T_c + \xi_r + e_{icr}$
- Effect on acceptance A among applicants (strata): $A_{ics} = \gamma_1 T_c + \beta X_i + \xi_s + e_{ics}$
- Hetergeneity by municipalities: $A_{ics} = \gamma_1 T_c + \gamma_2 T_c \times W_m + \gamma_3 W_m + \beta X_i + \xi_s + e_{ics}$

Higher wage $\rightarrow$ High ability applicants $\Uparrow$

	Observations (1)	Control (2)	Treatment effect (3)
Number of applicants	106	18.093	4.714 [4.430]
Panel A: Market skills			
Wage in previous job	1,572	3479.667	819.154 [174.703]***
Previous job was white collar	1,170	0.243	0.069 [0.029]***
Currently employed	2,225	0.104	0.053 [0.019]***
Has work experience	2,212	0.459	0.167 [0.048]***
Years of experience in past 3 spells	2,212	1.185	0.284 [0.171]
IQ (Raven test)	2,229	8.488	0.506 [0.223]**
Raven score ≥ 9	2,229	0.572	0.091 [0.039]**
Chose dominated risk option	2,213	0.431	−0.064 [0.025]**
Years of schooling	2,198	14.552	0.091 [0.308]

	Observations (1)	Control (2)	Treatment effect (3)
Panel B: Personality traits			
Extraversion	2,189	3.674	0.013 [0.036]
Agreeableness	2,167	4.107	0.004 [0.022]
Conscientiousness	2,191	4.235	0.063 [0.030]**
Neuroticism	2,168	2.254	−0.099 [0.033]***
Openness	2,168	3.910	0.042 [0.028]
Big 5 index	2,099	0.000	0.087 [0.049]*
Integrity: direct	2,223	0.067	−0.009 [0.013]
Integrity: indirect	2,099	44.424	0.602 [1.232]

Higher wage $\rightarrow$ Better motivated applicants $\Uparrow$

	Observations (1)	Control (2)	Treatment effect (3)
Panel A: PSM traits			
PSM index	2,074	0.000	0.092 [0.046]**
Attractiveness	2,217	2.803	0.070 [0.041]*
Commitment	2,170	3.316	0.045 [0.035]
Social justice	2,180	3.646	0.075 [0.026]***
Civic duty	2,158	3.924	0.027 [0.033]
Compassion	2,168	3.001	0.066 [0.031]**
Self-sacrifice	2,168	3.687	0.039 [0.034]
Panel B: Prosocial behavior			
Altruism	2,199	23.491	0.039 [0.291]
Negative reciprocity	2,206	0.508	0.075 [0.023]***
Cooperation	2,157	26.174	0.675 [0.404]*
Did charity work in the past year	2,223	0.605	−0.096 [0.041]**
Volunteered in the past year	2,224	0.710	−0.006 [0.027]
Importance of wealth	2,025	3.159	0.107 [0.087]
Belongs to a political party	2,225	0.113	−0.026 [0.014]*
Voted	2,225	0.758	0.019 [0.035]

Higher wage → Applicants more likely to accept

	Accepted (1)	Accepted (2)	Rejected (3)	Not reachable (4)
High-wage offer	0.151	0.160	−0.017	−0.135
	[0.054]***	[0.054]**	[0.034]	[0.054]***
Characteristics				
Male		−0.080		
		[0.058]		
Years of schooling		−0.022		
		[0.008]**		
High IQ		−0.017		
		[0.053]		
Wage in previous job > 5,000 pesos		−0.007		
		[0.067]		
Big 5 index		0.042		
		[0.038]		
PSM index		−0.024		
		[0.044]		
Mean of dependent variable	0.55	0.55	0.13	0.32
Observations	350	343	350	350
R-squared	0.10	0.12	0.09	0.13

Higher wage needed to offset disadvantageous regional traits

	Acceptance				
Dependent variable	(1)	(2)	(3)	(4)	(5)
High-wage offer	0.047	0.066	0.075	0.056	0.147
	[0.037]	[0.047]	[0.053]	[0.041]	[0.047]***
High wage offer × Distance	0.026			0.028	0.010
	[0.005]***			[0.007]***	[0.007]
Distance	−0.027			−0.028	−0.017
	[0.004]***			[0.007]***	[0.005]***
High wage offer × Drug-related deaths per 1,000 inhabitants		0.078		−0.033	−0.025
		[0.039]*		[0.037]	[0.040]
Drug-related deaths per 1,000 inhabitants		−0.107		0.000	−0.022
		[0.044]**		[0.045]	[0.040]
High wage offer × Human Development Index			−1.482	−0.913	−1.368
			[0.738]**	[0.645]	[0.589]**
Human Development Index			1.526	1.044	1.002
			[0.673]**	[0.598]*	[0.521]*
Observations	238	238	238	238	348
R-squared	0.25	0.21	0.2	0.26	0.18

Additional findings and takeaways

Labor supply elasticity (ϵ) for public sector with respect to wages

- $\epsilon = \epsilon_{\text{pool}} + \epsilon_{\text{accept}} + \epsilon_{\text{pool}} \times \epsilon_{\text{accept}} \times$ wage difference
- Wage difference: 33% (=(5000-3750)/3750)
- Pool: 4.7 person $\uparrow$ from 18.1 person (26.3%$\uparrow$) (Table 3) $\rightarrow$ Elasticity: 0.8 (=26.3/33)
- Acceptance: 15.1 p.p $\uparrow$ from 42.9% (35.2% $\uparrow$) (Table 5) $\rightarrow$ Elasticity: 1.07 (=35.2/33)
- $0.8 + 1.07 + 0.8 \times 1.07 \times 0.33 = 2.15$ (1% $\uparrow$ in wages $\rightarrow$ Supply 2.15% $\uparrow$)

Takeaways

- Offering high wages attracts better and motivated candidates
- However, findings hold only if motivation and ability are positively correlated
  - What if service commitment is sacrificed for higher pay (prioritize superiors over citizens?)
- Silent on post-acceptance performance: High pay to keep people motivated?

Desserano (2019):
Financial Incentives as Signals: Experimental Evidence from the Recruitment of Village Promoters in Uganda

Q: How does financial incentives shape perception on public posting?

What does financial incentives signal to applicants when informaion is incomplete?

- Information about the job is incomplete in many instances
- Financial incentives may provide signal on the nature of the task and the organization
  - High pay $\rightarrow$ More demanding and less enjoyable
  - May signal exploitative intentions
- How does this apply to public positions that require pro-social characteristics?

This paper tests the effect of fiscal incentives on job perceptions and application decisions

- Experiment with new position involving social and business task in Uganda
- Financial incentives signal the business-oriented nature
- Affects composition and size of the appicant pool
  - More applicants, but discourage applicants with pro-social preferences
- Thus, financial incentives may crowd out pro-social motives under incomplete info

Experimental setting

Recruitment of community health promoters (CHP) in Uganda

- Hire unskilled local agents and teach them how to serve their own villages
- Social component: Promote health education
- Business side: Sell household products at low prices
- Completely new: No prior on relative importance, difficulty, recruiter's goals etc

The experiment creates variations in expected earnings (fix actual incentives)

- Most CHPs talk together: Unequal actual incentives could cause problems
- Observe how differences in expectations affect types of applicants at selection
- How this was implemented
    - Applicants are told that earnings vary across workers (profit margin for selling)
    - Randomly given a different point of the income distribution (max, avg, min)
    - This defines high/medium/low-pay treatments

Experiment details

brac **Do you want to become COMMUNITY HEALTH PROMOTER?**

Only literate women are eligible

What will you do?

Educate your community on disease prevention and treatment	Provide your community with access to medicines and health products	Identify and assist pregnant women
		

Why should you become Community Health Promoter?

- Gain the skills to prevent diseases and promote health for your family and neighbours
- Serve your community
- Become a respected leader in your community

How much will you earn?

[Low-Pay, Medium-Pay or High-Pay Treatment Here]

How will you acquire health knowledge?

2 weeks of initial training + 1 day refresher each month

⇩

Learn about the most important health issues
+ Become highly trained on how to treat and prevent them

Low-Pay Treatment

How much will you earn?

- Buy medicines and health products from BRAC at a low price (panadol, fortified oil, soap, etc.)
- Sell them at a price higher than purchase price
- The more you sell, the more you earn

CHPs earn at least **7,000** Ushs per month

Medium-Pay Treatment

How much will you earn?

- Buy medicines and health products from BRAC at a low price (panadol, fortified oil, soap, etc.)
- Sell them at a price higher than purchase price
- The more you sell, the more you earn

CHPs earn an average of **30,000** Ushs per month

High-Pay Treatment

How much will you earn?

- Buy medicines and health products from BRAC at a low price (panadol, fortified oil, soap, etc.)
- Sell them at a price higher than purchase price
- The more you sell, the more you earn

CHPs earn up to **200,000** Ushs per month

Empirical strategy

Recruitment and information experiments

- Recruitment: Advertised to 4,863 eligible candidates in 315 communities
  - Treatment adminstered to microfinance groups and interviewed on intent to apply
- Information: 6,844 respondents asked to report their perception of the job
  - Gathered socioeconomic questions and pro-social preference information
  - 1/2 asked on their expected earnings, 1/2 asked on their perception
- Prosociality: Volunteer experience, and job feature the applicant prioritizes

Regression analysis

- Information treatment

$$Y_i = \alpha + \beta_1 \text{Medium}_i + \beta_2 \text{High}_i + X_i \eta + e_i$$

- Recruitment treatment: Interact Medium and High with pro-social traits
  - One version with microfinance group FE
  - Other with NGO branch FE (with dummy for trait, non-trait)

Financial incentives signal priority on private goals (vs. communal)

Variables	Job is perceived as a "private goal" (CHPs do the job for the money) more than a "social goal" (CHPs do the job to improve health conditions) (1)	Perceived proportion of time allocated to sales (versus delivery of health services) (2)	Expected number of work hours in a "typical week" (3)	Perceived difficulty in selling products to community (4)	Perceived difficulty in improving people's health behavior (5)	Own perceived ability (6)
Medium-pay treatment	−0.0033	0.0027	−0.0638	−0.0285	−0.0315	0.0286
	(0.02)	(0.01)	(0.34)	(0.04)	(0.05)	(0.13)
High-pay treatment	0.0692	0.0450	0.1665	0.0008	−0.0236	0.1144
	(0.02)	(0.01)	(0.35)	(0.04)	(0.05)	(0.14)
Mean dep. var. in low-pay treat.	0.403	0.461	14.081	1.827	2.536	6.004
Observations (number of respondents)	3,067	3,014	2,769	3,055	3,056	2,901
R^2	0.282	0.293	0.384	0.266	0.217	0.259
p-value Med = High	0.002	0.000	0.544	0.444	0.867	0.483
p-value Low = Med = High	0.004	0.000	0.822	0.698	0.789	0.673

Prosociality declines in high-pay treatment

(1)-(4): Microfinance group FE
(5)-(8): Branch FE

	Dependent variable = 1 if the potential candidate applies for the CHP position, = 0 if the potential candidate does not apply							
	Pro-social preferences		Interest in sales		Pro-social preferences		Interest in sales	
	Has ever volunteered in the health sector	Community driven	Owns a shop	Has ever sold health-related products	Has ever volunteered in the health sector	Community driven	Owns a shop	Has ever sold health-related products
TRAIT	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
TRAIT	0.2812	0.0865	0.0271	0.0503	0.2749	0.0703	0.0302	0.0834
	(0.05)	(0.02)	(0.03)	(0.04)	(0.04)	(0.02)	(0.03)	(0.04)
TRAIT × Medium-pay treatment	−0.0017	−0.0441	0.0233	0.1156	−0.0121	−0.0073	0.0188	0.0710
	(0.06)	(0.03)	(0.04)	(0.06)	(0.06)	(0.02)	(0.04)	(0.06)
TRAIT × High-pay treatment	−0.1621	−0.0882	0.0566	0.1587	−0.1078	−0.0139	0.0738	0.1318
	(0.06)	(0.03)	(0.05)	(0.06)	(0.06)	(0.02)	(0.04)	(0.06)
NOT_TRAIT × Medium-pay treatment					0.0197	0.0326	0.0167	0.0114
					(0.01)	(0.02)	(0.02)	(0.01)
NOT_TRAIT × High-pay treatment					0.0550	0.0768	0.0320	0.0303
					(0.02)	(0.02)	(0.02)	(0.02)

Other findings and remaining questions

Other findings: Financial incentives may crowd-out pro-social preferences

- The size of the pool is largest in high-pay treatment
- However, those recruited under high-pay remain less pro-social even after working
- Higher retention of CHP workers in low-pay treatment

Remaining questions

- Financial incentives highlights business aspects of the work, even after recruitment
  - Any way to enhance pro-social motives afterwards?
- Different setting compared to Dal Bó, Finan, Rossi (2013)
  - Known job of public workers vs uncertain characteristics of CHPs
  - Financial incentives may not have a signalling effect in the former
  - Thus, the crowding-out effect may not be observed in some cases

Moreira, Pérez (2024):
Civil Service Exams and Organizational Performance: Evidence from the Pendleton Act

Q: How do civil service exams affect organizational performance?

Professionalized bureaucracy increasingly seen as key for economic development

- Many tries to achieve this by Introducing competitive exams
- The intention is to limit favoritism and help select more qualified empoyees
- But additional positions not covered by exams could distort hierarchical structure
- So the effect on performance may be ambiguous event with better selection

This paper studies impact of civil service exams on organizational performance in the US

- Using 1883 Pendleton Civil Service Reform Act (mandated exams for federal workers)
- Focus on Customs Service (responsible for large share of federal revenues)
- Despite improved selection and retention, organizational performance did not improve
- Increases in exam-exempted positions and retentions of managers explains findings

Historical Background

Customs revenue collection in historic US

- Customs revenue accounted for more than half the federal revenues in 1880s
- Hiring was based on political and personal connections prior to reforms
- Inefficient practies: Cost of collecting $1 larger in US than DEU, ENG, FRA
  - Largely due to overstaffing and 'errors and frauds' of these employees

Implementation of the 1883 Pendleton reforms on customs

- Applied to districts with 50+ employees and positions earning $900+ by 1883
  - These districts are called 'classified' positions
  - Only allowed to select candidates among top 4 scores in open exams
- Tested on practical knowledge relevant to the position rather than academic training
- Other components applied to ALL districts (making treated vs control comparable)
  - No political assessments and hiring individuals 'using intoxicating bevarages to excess'

Data and empirical strategy

Data sources: Covers 1871-1893 (Authors digitized the records themselves)

- Personnel data: Official Registers of the US (biennial) linked to US population census
  - Name, job title, state/country of birth, compensation, place of employment
- Finance records: Annual Report of Secretary of the Treasury on State of the Finances
  - Revenues and expenditures of different segments of government agencies

Leverage difference-in-differences induced by 1883 reforms

$$Y_{ct} = \alpha_c + \alpha_t + \beta \text{Classified}_c \times \text{After}_t + \gamma X_{ct} + e_{ct}$$

- In some regressions, β is broken down across different time periods
- Authors verify that outcomes evolve in parallel pattern pre-reform (parallel trends)
- No evidence of districts manipulating their size downward to avoid reforms

Exams $\rightarrow$ Hiring of those with professional backgrounds

	All		New hires		New hires, no exam		New hires, exam	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A. Professional occupation in census								
Classified × *After*	−0.00343	0.00381	0.0220	0.0210	−0.0560	−0.0604	0.0740	0.0730
	(0.0195)	(0.0167)	(0.0244)	(0.0231)	(0.0401)	(0.0400)	(0.0283)	(0.0284)
District FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Region × *After* FE	No	Yes	No	Yes	No	Yes	No	Yes
Observations	8,614	8,614	3,051	3,051	1,018	1,018	2,033	2,033
Clusters	46	46	46	46	46	46	45	45
Mean of dep. var.	0.0723	0.072	0.0757	0.0757	0.0629	0.0629	0.0821	0.0821
p-value							0.00688	0.00939
Panel B. Unskilled or no occupation in census								
Classified × *After*	−0.0347	−0.0382	−0.0437	−0.0440	−0.0207	−0.0261	−0.0716	−0.0714
	(0.0281)	(0.0257)	(0.0402)	(0.0406)	(0.0656)	(0.0682)	(0.0428)	(0.0406)
District FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Region × *After* FE	No	Yes	No	Yes	No	Yes	No	Yes
Observations	8,614	8,614	3,051	3,051	1,018	1,018	2,033	2,033
Clusters	46	46	46	46	46	46	45	45
Mean of dep. var.	0.202	0.202	0.236	0.236	0.280	0.280	0.213	0.213
p-value							0.432	0.485

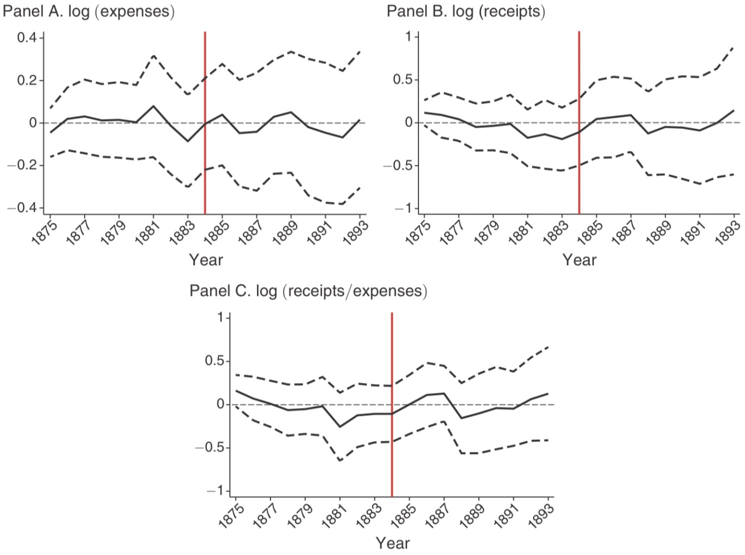
Less likely to leave after reform

	All		No exam		Exam	
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A. Positions exempted from exams versus positions non exempted						
Classified × *After*	−0.116	−0.109	−0.0723	−0.0783	−0.144	−0.138
	(0.0273)	(0.0302)	(0.0465)	(0.0464)	(0.0258)	(0.0291)
District FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Region × *After* FE	No	Yes	No	Yes	No	Yes
Observations	41,680	41,680	10,108	10,108	31,572	31,572
Clusters	46	46	46	46	46	46
Mean of dep. var.	0.371	0.371	0.470	0.470	0.338	0.338
p-value					0.0515	0.0972

	No transition		Transition	
	(1)	(2)	(3)	(4)
Panel B. Years with presidential transitions versus years without presidential transitions				
Classified × *After*	−0.0654	−0.0723	−0.205	−0.180
	(0.0336)	(0.0360)	(0.0567)	(0.0502)
District FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Region × *After* FE	No	Yes	No	Yes
Observations	22,864	22,864	18,816	18,816
Clusters	46	46	46	46
Mean of dep. var.	0.331	0.331	0.408	0.408
p-value			0.0529	0.0811

Signals that exams reduced the role of political alignment and favoritism in allocating jobs

However, reforms did not lead to better cost-effectiveness



Mechanisms and unanswered questions

The incomplete reach of the reforms

- Managers and positions with salaries below a threshold not subject to reforms
- The number of low-paid positions inflated following reforms (twice)
- Selection of managers were left unchanged

Room for further research on other aspects

- Reducing favoritism in hiring can increase retention and quality in selection
- But does not automatically lead to better practices
  - Need to prevent responses that can offset the effects of reforms
  - Managers also matter - Need to minimize incomplete reach
- How to design organizational reform that minimizes these side effects?

Remaining questions for future research

Attracting capable and intrinsically-motivated individuals

- Higher wages attract capability but may crowd out pro-social preferences
- Non-monetary incentives matter but hard to design at scale

When does discretion improve selection vs. open the door to patronage?

- Optimal mix of rule-based and discretionary selection likely varies by context
- How do political institutions shape this tradeoff?

Does better selection translate into better performance?

- Selection is only one part: Need to ensure employees are incentivized afterwrads
- How to design organizational structure that maximizes incentives post-selection?